Aishwarya Mudgal Sunil Kumar

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Experience

Viavi Solutions Inc | Hopkins, Minnesota
Software Development Engineer II *May 2026 – Present*

- Drove UI development for Observer Apex's Threat Hunting / Threat Intelligence line — IP threat indicators, Threat Forensics dashboards, reconstruction workflows, and a timeline visualization built on existing Apex widgets
- Designed a reusable threat-indicator and shipped it across graphs, tables and PDF/CSV exports, with click-to-drilldown access into IOC threat data
- Built alert investigation status/history tracking and CSV export for compliance reporting with filter-aware streaming

Software Development Engineer I *August 2023 – May 2026*

- Owned the Network Segmentation Tags feature end-to-end — API design, CRUD, import/export
- Built a reusable folder-based organization system for dashboards and widgets — creation, hierarchy view, expand/collapse, and import/export that preserves folder assignment
- Migrated the frontend unit test suite from Karma to Jest and refactored legacy jQuery components to React

Ensco, Inc – Rail | Charlottesville, Virginia *May 2019 – May 2021*
Software Engineer I

- Built a cross-platform application in Angular for the Federal Railroad Administration to assess the safety of railway tracks
- Created custom, responsive, and interactive charts using Chart.js to show probability of buckling of railroads
- Developed features such as reactive forms with validations, modal pop-ups, and user authentication with Router Guards

Skills

Programming: JavaScript, TypeScript, HTML, CSS, Python, Java, C#, SQL
Data Analysis and Visualization: D3.js, Tableau, Chart.js, R (ggplot2), Vega-Lite, Scikit-learn, SciPy, Pandas, Excel
Front-End Development: React, jQuery, Angular, Svelte, Bootstrap
Prototyping Tools: Figma, ProtoPie, Axure RP
AI Tools: Claude

Education

Georgia Institute of Technology | Atlanta, GA *August 2021 – May 2023*
Master of Science in Computer Science (Visual Analytics – Project track)

Purdue University | West Lafayette, IN *August 2014 – December 2018*
Bachelor of Science in Computer Science (Machine Learning)

Research/Publications

Georgia Institute of Technology | Atlanta, Georgia *January 2022 – May 2023*
Visualization Recommendation Systems / Advised by Dr. Alex Endert

- Designed and developed VisTA (Visualizations for Tutor Analytics), a visual analytics tool that helps teachers interpret student interaction data from an intelligent tutoring system and identify necessary follow-up actions

*Guo, G., **Mudgal Sunil Kumar, A.**, Gupta, A., Coscia, A., MacLellan, C., & Endert, A. (2024). **Visualizing Intelligent Tutor Interactions for Responsive Pedagogy.** In Proceedings of the 2024 ACM International Conference on Advanced Visual Interfaces (AVI '24). Association for Computing Machinery. (arXiv)*

Projects

Multi-Touch based Data Exploration tool with D3.js *(View)* *September 2022 – December 2022*
A data exploration tool using touch/pointers – optimized for tablets (Group project — CS 6456: Principles of UI Software)

- Built a data exploration tool based on unit visualizations that allows for touch/pointer-based interactivity
- Features include lasso selection, grouping, zooming/panning, sorting, updating data cases with colors, shapes, images etc.